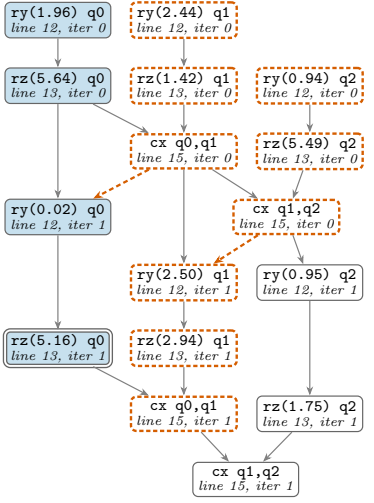
(a) VQE ansatz

```
n = 3
layers = 2
rng = np.random.default_rng(7)

qc = QuantumCircuit(n, n)

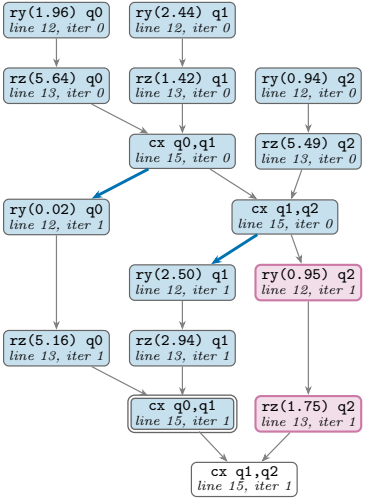
for layer in range(layers):
    for qubit in range(n):
        qc.ry(float(rng.uniform(0, np.pi)), qubit)
        qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
    for qubit in range(n - 1):
        qc.cx(qubit, qubit + 1)
```

(b) Classical PDG



slice of q_0: 4 of 16 operations
fidelity 0.77, misses 9 dependences

(c) Wire rule



slice of q_0: 13 of 16 operations
fidelity 1.00, exactly the dependences

data · quantum · quantum edge missing from the classical PDG · in the slice · dependence the slice misses
added by entanglement closure, not a dependence · last write to the criterion